

IBM BOB HACKATHON 2026

# PR Surgeon

Decompose monster Pull Requests into safe, reviewable sub-PRs.

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Powered by IBM Bob · watsonx.ai Granite

Theme: Turn idea into impact faster · Solo build, 18 hours

## THE PROBLEM

# Enterprise migrations create monster PRs

When a bank migrates Angular to React, a telco moves Java 8 to 17, or a monolith splits into microservices, the result is the same: one Pull Request with 300-500 files changed

- **No one can hold the full diff**  
Senior engineers need 6+ hours per 300-file PR and still miss cross-file regressions.
- **It blocks the team for 4-8 weeks**  
The whole team works around merge conflicts and feature freezes while it sits open.
- **It ships subtle regressions**  
Rushed reviews convert into production bug debt.

**\$15-25K**

cost per monster PR

**50-200 / yr**

per Fortune 500 eng org

(DORA, 2024)

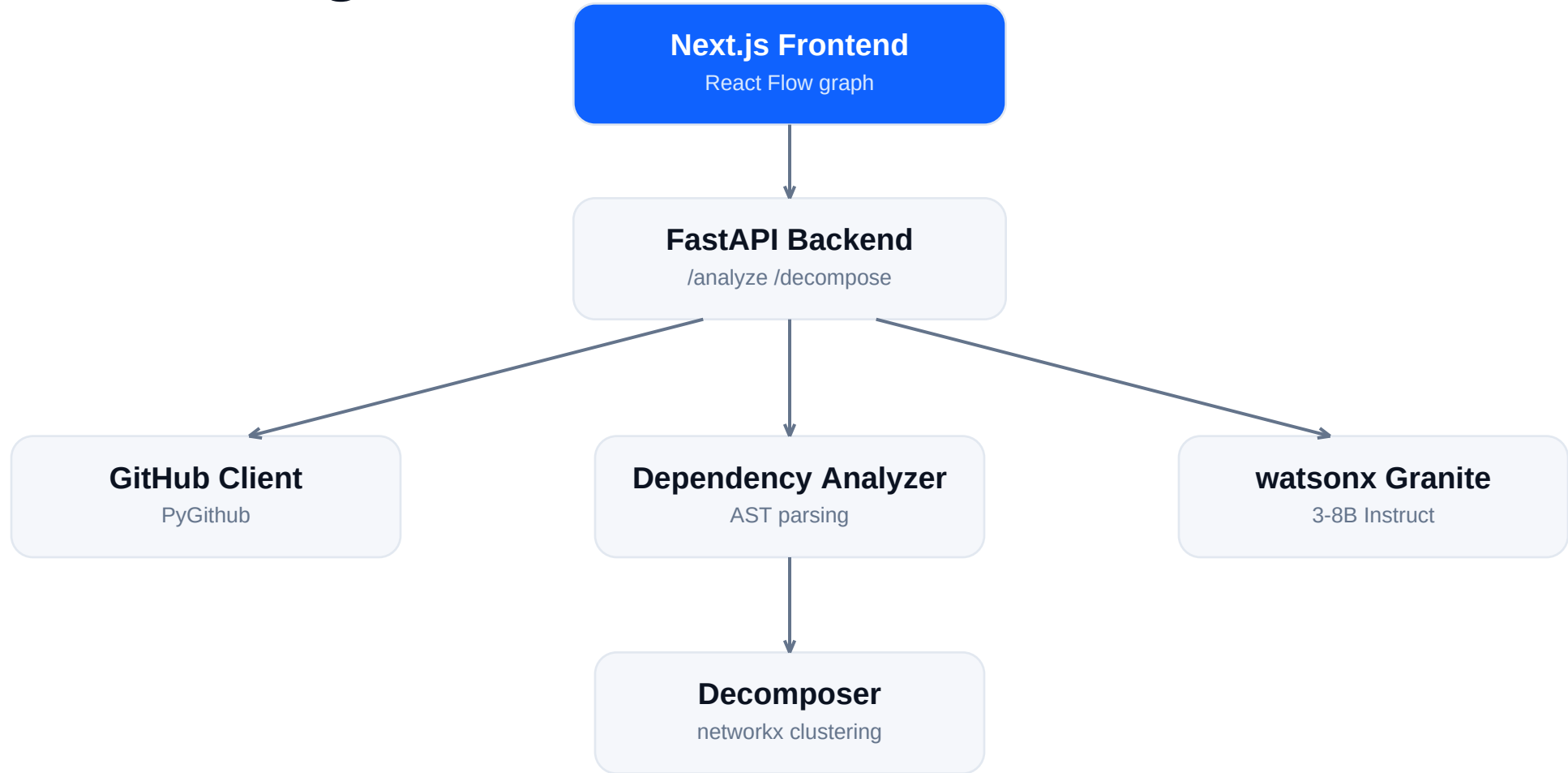
## THE SOLUTION

# Paste a PR URL, get a safe split plan



**A 4-8 week review marathon becomes a 5-10 day mergeable plan.**

# System design



# Built to ship in 18 hours

## Backend

- Python 3.11
- FastAPI
- PyGithub
- networkx
- Pydantic v2

## Frontend

- Next.js 14
- TypeScript
- React Flow
- shadcn/ui
- Tailwind

## AI

- IBM watsonx.ai
- Granite 3-8B
- Template fallback

## Deploy

- Railway (API)
- Vercel (web)

WHY ONLY IBM BOB

# This needs whole-repo context

Decomposing a PR is not a single-file task. The tool must reason that `models/user.py` is consumed by `routes/auth.py`, that a React form calls a backend route, and that migrating `user_id` int -> uuid cascades through 47 files.

A snippet-completion LLM cannot do this. Bob already understood the import graph from `/init`, so it proposed community-detection grounded in real graph topology - not generic CS advice. This is the use case Bob was built for.

**Full repo context is the foundation that makes the analysis trustworthy.**

## HOW BOB WAS USED

# Seven sessions, one build

| Session                | Phase      | Outcome                     |
|------------------------|------------|-----------------------------|
| 01 Scaffolding         | Hour 0-2   | Monorepo, AGENTS.md, config |
| 02 GitHub client       | Hour 2-4   | PyGithub wrapper + tests    |
| 03 Dependency analyzer | Hour 4-7   | Graph logic, 3 iterations   |
| 04 Decomposer          | Hour 7-9   | Topo order + cluster sizing |
| 05 watsonx integration | Hour 9-11  | Granite prompts + fallback  |
| 06 Frontend graph      | Hour 11-13 | React Flow component        |
| 07 Polish & docs       | Hour 14-16 | README, submission, script  |

Bobcoin used: ~30 / 40 (75%) · Custom "Algorithm Design" mode for the analyzer

# A force multiplier for modernization

**\$15-25K**

saved per processed PR  
(vs. consulting rates)

**TAM ~10,000 orgs**

SAM ~2,000 modernizing  
SOM yr-1: 50 customers

**Pricing paths**

\$99 per PR analyzed  
\$5K/mo enterprise  
IBM Consulting white-label

The white-label path plugs directly into IBM Garage / IBM Consulting modernization engagements - the highest-value route.



# First to auto-propose the split

- **Other hackathon teams target different spaces**  
EDA, CV, ticketing, compliance - none address migration PR decomposition.
- **GitHub's "split this PR" is manual**  
Checkbox-driven, zero intelligence.
- **git split-diffs & friends are viewers**  
They show diffs; they do not decompose.
- **Graphite / stacked-PR tools assume you already know the split**  
They help execute a split you designed by hand.

**PR Surgeon is the first tool that uses full-repo-context AI to propose the split.**

# Working today, honest about cuts

## Shipped

- Backend pipeline complete (56 tests passing)
- 8 sub-PRs generated from a 43-file PR
- Python AST + JS/TS regex parsing
- Frontend running, dark mode, Vercel-ready
- All 7 Bob sessions exported as evidence

## Next (v2)

- GitHub App: inline split suggestions
- Auto-create the sub-PR branches
- GitLab / Bitbucket support
- Go, Java, Rust, C# parsers
- Tree-sitter for JS/TS accuracy

THANK YOU

# From idea to impact, faster.

PR Surgeon turns the most expensive artifact in enterprise software - the monster migration PR - into a safe, reviewable plan. Built in 18 hours, solo, with IBM Bob as the partner.

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